USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [23ESME12]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY- 2025

ELEMENTS OF MECHANICAL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART-A	**5 X 2=10**		
1.	**a**	List the difference between impulse and reaction turbine.	L 2	CO 1	**02**
	b.	What is a gear? List the types of gears.	L 2	CO 2	**02**
	c.	What is lubrication? State the role of lubrication in bearings.	L 2	CO 2	**02**
	d.	Give comparison between temporary joining and permanent joining.	L 2	CO 3	**02**
	e.	List the different types of lathe machines.	L 2	CO 4	**02**
		PART-B	**6X5=30**		
2.	**a.**	Explain with neat sketch the working principle of hydroelectric power plant.	L 2	CO 1	**05**
		OR			
	b.	Explain with neat sketch the working principle of nuclear power plant.	L 2	CO 1	**05**
3.	**a.**	What is centrifugal pump? With a neat sketch explain the working principle of centrifugal pump.	L 2	CO 1	**05**
		OR			
	b.	Explain the applications of springs. State the materials used in making the springs.	L 2	CO 2	**05**
4.	**a.**	What is welding? Name the different types of welding joints with neat sketches.	L 2	CO 3	**05**
		OR			
	b.	Explain with neat sketch worm gear.	L 2	CO 2	**05**
5.	**a.**	Explain with neat sketch simple gear train.	L 2	CO 2	**05**
		OR			
	b.	Explain in detail with neat sketch the classification of helical springs.	L 2	CO 3	**05**
6.	**a.**	Explain in detail geometric dimensioning and tolerance.	L 3	CO 5	**05**
		OR			
	b.	Explain in detail caulking process and with a neat sketch.	L 2	CO 3	**05**
7.	**a.**	Explain with neat sketch the working principle of milling machine.	L 2	CO 4	**05**

Jain Deemed-to-be University Learning Resource Centre Jain Global Campus Ramanagar-112

		OR			
	b.	What is a machine tool? Explain the basic elements of machine tools.	L 2	CO 6	**05**
		PART-C	**6X10=60**		
8.	**a.**	Explain with neat sketch the construction and working principle of Francis turbine. State it's advantages and disadvantages.	L 2	CO 1	**10**
		OR			
	b.	Explain with neat sketch the working principle of tidal power plant. State it's advantages and disadvantages.	L 2	CO 1	**10**
9.	**a.**	Explain with neat sketch the basic components of wind power plant. State it's advantages and disadvantages.	L 2	CO 1	**10**
		OR			
	b.	Explain in detail limits, fits and tolerances with suitable examples.	L 3	CO 5	**10**
10.	**a.**	Explain with neat sketch the layout and working principle of thermal power plant. State it's advantages and disadvantages.	L 2	CO 1	**10**
		OR			
	b.	A pinion gear with 22 teeth and module of 6mm has a rotational speed of 1200rpm and drives a gear at 660rpm. Determine: **i.** Number of teeth on gear. **ii.** Center distance.	L 2	CO 2	**10**
11.	**a.**	Explain in detail with neat sketch the classification of bearing based on the type of friction.	L 2	CO 2	**10**
		OR			
	b.	Explain the different types of helical springs and their applications in mechanical systems	L 2	CO 2	**10**
12.	**a.**	Explain in detail the different types of Rivets with neat sketches for each. State the advantages, disadvantages and applications of rivet joints.	L 2	CO 4	**10**
		OR			
	b.	Explain in detail the three types of fits with neat sketches.	L 3	CO 5	**10**
13.	**a.**	What is a lathe machine? Explain with neat sketch the working principle of lathe machine. Give the classification of different operations of lathe machines.	L 2	CO 4	**10**
		OR			
	b.	What is milling machine? Explain with neat sketch the column and knee type of milling machine. State the operations of milling machine.	L 2	CO 4	**10**